Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech | ***Time: 03 Hours***
Code: 101505 | ***Subject: Hydrology & Water Resources Engineering*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The mean annual precipitation over a catchment is 1100 mm. If the mean annual runoff is 450 mm, the annual water loss due to evaporation and other abstractions is
(i) 1550 mm
(ii) 650 mm •
(iii) 1100 mm
(iv) 450 mm

(b) The pan coefficient for a Class A evaporation pan is 0.7. If the measured pan evaporation is 100 mm, the estimated lake evaporation is
(i) 70 mm •
(ii) 100 mm
(iii) 143 mm
(iv) 130 mm

(c) The SCS-CN method primarily estimates
(i) Peak discharge
(ii) Total runoff volume •
(iii) Base flow
(iv) Rainfall intensity

(d) For a fully penetrating well in a confined aquifer, the steady-state discharge is proportional to
(i) The square of the drawdown
(ii) The product of transmissivity and drawdown •
(iii) The difference in hydraulic head only
(iv) The cube of the well radius

(e) According to Lacey's regime theory, the silt factor 'f' is a function of
(i) Channel slope
(ii) Silt size •
(iii) Mean velocity
(iv) Channel width

(f) The elementary profile of a gravity dam is designed such that
(i) The resultant force passes through the middle third of the base •
(ii) The base width is minimized
(iii) The maximum compressive stress at the toe is zero
(iv) The resultant acts at the heel

(g) The depth-area-duration (DAD) curve is used to determine
(i) Peak rainfall intensity •
(ii) Rainfall frequency
(iii) Areal distribution of rainfall
(iv) ain gauge calibration

(h) The Blaney-Criddle method is used to estimate
(i) Infiltration capacity
(ii) Potential evapotranspiration •
(iii) Pan evaporation
(iv) Depression storage

(i) The unit hydrograph is defined as the direct runoff hydrograph resulting from
(i) 1 mm of rainfall excess in 1 hour
(ii) 1 cm of rainfall excess in a specified duration •
(iii) 1 mm of rainfall over the catchment
(iv) 1 cm of rainfall in 1 hour

HIGHLY CONFIDENTIAL DOCUMENT EDC 111 @ 25-07-2025 09:42:19
© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

(j) Water logging in irrigated lands is mainly caused by
(i) Excessive canal lining
(ii) High permeability of soil
(iii) Inadequate surface drainage
(iv) Low groundwater table

Q.2 (a) Explain the hydrologic cycle and its significance in water resources engineering. [7]
(b) Describe the different types of precipitation and their measurement techniques. [7]

Q.3 (a) Discuss the various methods of evaporation estimation and factors affecting evaporation from water bodies. [7]
(b) Class A pan records 120 mm evaporation in a week. If the pan coefficient is 0.7, estimate the actual evaporation from a nearby lake. Also, discuss the limitations of pan evaporation data. [7]

Q.4 (a) Explain the SCS-CN method for estimation of runoff. What are its main assumptions and limitations? [7]
(b) Given a catchment with Curve Number (CN) = 75 and a rainfall event of 80 mm, estimate the direct runoff using the SCS-CN method. (Assume initial abstraction = 0.2S, where S = (25400/CN) – 254) [7]

Q.5 (a) Explain the difference between confined and unconfined aquifers with neat sketches. [7]
(b) Derive the expression for steady-state discharge from a fully penetrating well in a confined aquifer. [7]

Q.6 (a) What is meant by 'duty' and 'delta' in irrigation? How are these terms related? [7]
(b) A crop requires a total depth of 60 cm of water during its base period of 120 days. Calculate the duty and delta for the crop. [7]

Q.7 (a) Discuss Kennedy's theory of channel design. What are its main assumptions and limitations? [7]
(b) A canal is to be designed to carry a discharge of 50 m^3/s using Lacey's theory. If the silt factor is 1.0, calculate the regime velocity and wetted perimeter. [7]

Q.8 (a) Describe the causes and effects of water logging. Suggest remedial measures. [7]
(b) Explain the design considerations for an earthen (embankment) dam and the methods to control seepage through the body of the dam. [7]

[3.5 x 4 = 14]

Q.9 Write short notes on ***any four*** of the following:
(a) Rain gauge network and its optimization
(b) Infiltration indices
(c) Flow duration curve
(d) Consumptive use of water
(e) Canal lining – types and advantages
(f) Spillway gates and their functions

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play